微積分期中考前複習

1 關於期中考

2 內容重點

關於期中考

考試時間：110 分鐘（17:10 — 19:00）

出題型態：計算題、證明題

英文考題；作答英文中文皆可

答題必須寫出（合理的）過程，不可只寫答案；答案對但解題思路觀念錯誤，一樣酌情扣分

考試範圍：至 3.3 節（包含）

必考題：

極限 ε–δ 定義敘述

導數、微分定義敘述

指定習題（至 3.3 節共 15 題）為「高機率」考題

內容重點

極限 ε–δ 定義
導數、微分定義
微分規則：乘除微分、鏈鎖律
隱微分
微分：三角函數、自然指數及對數函數
對數微分

極限 ε–δ 定義

State the formal ε–δ definition of $\lim_{x \to a} f(x) = L$.

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Given $\varepsilon > 0$,

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Given $\varepsilon > 0$, $\exists\, \delta > 0$ such that

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Given $\varepsilon > 0$, $\exists\, \delta > 0$ such that $|x - a| < \delta$

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State the formal ε–δ definition of $\lim_{x\to a} f(x) = L$.

Given $\varepsilon > 0$, $\exists\, \delta > 0$ such that $|x - a| < \delta \implies |f(x) - L| < \varepsilon$.

極限 ε–δ 定義

Use the formal ε–δ definition of limit to prove: if $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$ then $\lim_{x\to a} (f(x) + g(x)) = L + M$.

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$$|f(x) + g(x) - (L + M)|$$

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Take $\delta = \min\{\delta_1, \delta_2\}$, then for $|x - a| < \delta$,

$$|f(x) + g(x) - (L + M)| = |(f(x) - L) + (g(x) - M)|$$

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$$\begin{aligned} |f(x) + g(x) - (L + M)| &= |(f(x) - L) + (g(x) - M)| \\ &\leqslant |(f(x) - L)| + |(g(x) - M)| \end{aligned}$$

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Take $\delta = \min\{\delta_1, \delta_2\}$, then for $|x - a| < \delta$,

$$\begin{aligned} |f(x) + g(x) - (L + M)| &= |(f(x) - L) + (g(x) - M)| \\ &\leqslant |(f(x) - L)| + |(g(x) - M)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

導數、微分定義

State the formal definition of $f'(x)$ given differentiable $f(x)$.

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Formal definition of $f'(x)$ from $f(x)$:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

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Use the formal definition of derivatives to derive $f'(x)$ for $f(x) = x^{\frac{1}{3}}$.

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\begin{aligned}
f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \to 0} \frac{(x+h)^{\frac{1}{3}} - x^{\frac{1}{3}}}{h}
\end{aligned}
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&= \lim_{h \to 0} \frac{(x+h)^{\frac{1}{3}} - x^{\frac{1}{3}}}{h} \\
&= \lim_{h \to 0} \frac{(x+h) - x}{h\left((x+h)^{\frac{2}{3}} + (x+h)^{\frac{1}{3}} x^{\frac{1}{3}} + x^{\frac{2}{3}}\right)}
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&= \lim_{h\to 0} \frac{1}{\left((x+h)^{\frac{2}{3}} + (x+h)^{\frac{1}{3}}x^{\frac{1}{3}} + x^{\frac{2}{3}}\right)}
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&= \frac{1}{3x^{\frac{2}{3}}}
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微分規則：乘除微分、鏈鎖律

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$$(fg)' = f'g + fg'$$

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鏈鎖律（The Chain Rule）：

$$(f(g(x))' = f'(g(x))\,g'(x)$$

微分規則：乘除微分、鏈鎖律

Find $(f^{-1})'(-2)$ if $f(x) = x\sqrt{3+x^2}$.

微分規則：乘除微分、鏈鎖律

Find $(f^{-1})'(-2)$ if $f(x) = x\sqrt{3+x^2}$.

From the definition of inverse function,

$$f^{-1}(f(x)) = x$$

Differentiate above with x, we have

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$$\begin{aligned}(f^{-1}(f(x)))' &= 1\\ (f^{-1})'(f(x))f'(x) &= 1\\ (f^{-1})'(f(x)) &= \frac{1}{f'(x)}\end{aligned}$$

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Solving $x\sqrt{3+x^2} = -2$, we have $x = -1$; that is, $f(-1) = -2$.

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$$(f^{-1})'(-2) = (f^{-1})'(f(-1)) = \frac{1}{f'(-1)} = \frac{2}{5}$$

微分規則：乘除微分、鏈鎖律

Find $(f^{-1})'(0)$ for $f(x) = \frac{1}{2} + \sin x$, $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

微分規則：乘除微分、鏈鎖律

Find $(f^{-1})'(0)$ for $f(x) = \frac{1}{2} + \sin x$, $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

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Solving $\frac{1}{2} + \sin x = 0$, we have $\sin x = -\frac{1}{2}$; for $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, $x = -\frac{\pi}{6}$. that is, $f(-\frac{\pi}{6}) = 0$.

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微分規則：乘除微分、鏈鎖律

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$$(f^{-1})'(0) = (f^{-1})'(f(-\tfrac{\pi}{6})) = \frac{1}{f'(-\frac{\pi}{6})} = \frac{1}{\cos\left(-\frac{\pi}{6}\right)} = \frac{2}{\sqrt{3}}.$$

隱微分

將 y 看作為 x 的函數（$y \equiv y(x)$）；鏈鎖律

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將 y 看作為 x 的函數 $(y \equiv y(x))$；鏈鎖律

$$(xy)' =$$

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$$(e^{xy^2})' =$$

隱微分

將 y 看作為 x 的函數 $(y \equiv y(x))$；鏈鎖律

$$(xy)' = xy' + y$$
$$(e^{xy^2})' = e^{xy^2}(y^2 + 2xyy')$$

微分：三角函數、自然指數及對數函數

三角函數

$$(\sin x)' =$$

自然指數及對數函數

微分：三角函數、自然指數及對數函數

三角函數

$$(\sin x)' = \cos x$$

自然指數及對數函數

微分：三角函數、自然指數及對數函數

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$$(\sin x)' = \cos x$$
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$$(\cos x)' = -\sin x$$

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三角函數

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$$\lim_{h \to 0} \frac{\sin h}{h} =$$

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三角函數

$$(\sin x)' = \cos x$$
$$(\cos x)' = -\sin x$$
$$\lim_{h\to 0} \frac{\sin h}{h} = 1$$

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$$\lim_{h\to 0} \frac{e^h - 1}{h} =$$

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$$\lim_{h\to 0} \frac{e^h - 1}{h} = 1$$

$$(e^x)' =$$

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$$(e^x)' = e^x$$

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$$\lim_{h\to 0} \frac{e^h - 1}{h} = 1$$

$$(e^x)' = e^x$$

$$(a^x)' =$$

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自然指數及對數函數

$$\lim_{h\to 0} \frac{e^h - 1}{h} = 1$$

$$(e^x)' = e^x$$

$$(a^x)' = a^x \log a$$

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自然指數及對數函數

$$\lim_{h\to 0} \frac{e^h - 1}{h} = 1$$

$$(e^x)' = e^x$$

$$(a^x)' = a^x \log a$$

$$(\log x)' =$$

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自然指數及對數函數

$$\lim_{h\to 0} \frac{e^h - 1}{h} = 1$$

$$(e^x)' = e^x$$

$$(a^x)' = a^x \log a$$

$$(\log x)' = \frac{1}{x}$$

對數微分

複雜表示式（相乘、指數）型，先取對數

對數微分

複雜表示式（相乘、指數）型，先取對數

$$\frac{\mathrm{d}}{\mathrm{d}x}\log f(x) = \frac{f'(x)}{f(x)}$$

對數微分

Find $f'(x)$ for

$$f(x) = x^{e^x}$$

對數微分

Find $f'(x)$ for

$$f(x) = x^{e^x}$$

$$\log f(x) = e^x \log x$$

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Find $f'(x)$ for

$$f(x) = x^{e^x}$$

$$\log f(x) = e^x \log x$$

$$\frac{f'(x)}{f(x)} = e^x \log x + e^x \frac{1}{x}$$

對數微分

Find $f'(x)$ for

$$f(x) = x^{e^x}$$

$$\log f(x) = e^x \log x$$

$$\frac{f'(x)}{f(x)} = e^x \log x + e^x \frac{1}{x}$$

$$f'(x) = x^{e^x} e^x \left(\log x + \frac{1}{x} \right)$$

對數微分

Find $f'(x)$ for

$$f(x) = (\sin x)^{x^3}$$

對數微分

Find $f'(x)$ for

$$f(x) = (\sin x)^{x^3}$$

$$\log f(x) = x^3 \log(\sin x)$$

對數微分

Find $f'(x)$ for

$$f(x) = (\sin x)^{x^3}$$

$$\log f(x) = x^3 \log(\sin x)$$

$$\frac{f'(x)}{f(x)} = 3x^2 \log(\sin x) + x^3 \frac{1}{\sin x} \cos x$$

對數微分

Find $f'(x)$ for

$$f(x) = (\sin x)^{x^3}$$

$$\log f(x) = x^3 \log(\sin x)$$

$$\frac{f'(x)}{f(x)} = 3x^2 \log(\sin x) + x^3 \frac{1}{\sin x} \cos x$$

$$f'(x) = (\sin x)^{x^3} \left(3x^2 \log(\sin x) + x^3 \frac{1}{\sin x} \cos x \right)$$

對數微分

Find $f'(x)$ for

$$f(x) = \frac{(x+2)^3(2x+1)^9}{x^8(3x^2+1)^4}$$

對數微分

Find $f'(x)$ for

$$f(x) = \frac{(x+2)^3(2x+1)^9}{x^8(3x^2+1)^4}$$

$$\log f(x) = 3\log(x+2) + 9\log(2x+1) - 8\log x - 4\log(3x^2+1)$$

對數微分

Find $f'(x)$ for

$$f(x) = \frac{(x+2)^3(2x+1)^9}{x^8(3x^2+1)^4}$$

$$\log f(x) = 3\log(x+2) + 9\log(2x+1) - 8\log x - 4\log(3x^2+1)$$

$$\frac{f'(x)}{f(x)} = 3\frac{1}{x+2} + 9\frac{2}{2x+1} - 8\frac{1}{x} - 4\frac{6x}{3x^2+1}$$

對數微分

Find $f'(x)$ for

$$f(x) = \frac{(x+2)^3(2x+1)^9}{x^8(3x^2+1)^4}$$

$$\log f(x) = 3\log(x+2) + 9\log(2x+1) - 8\log x - 4\log(3x^2+1)$$

$$\frac{f'(x)}{f(x)} = 3\frac{1}{x+2} + 9\frac{2}{2x+1} - 8\frac{1}{x} - 4\frac{6x}{3x^2+1}$$

$$f'(x) = \frac{(x+2)^3(2x+1)^9}{x^8(3x^2+1)^4}\left(\frac{3}{x+2} + \frac{18}{2x+1} - \frac{8}{x} - \frac{24x}{3x^2+1}\right)$$